

The Complete Treatise on Extremum-Normalized Stochastic Process Theory

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Abstract

This treatise develops Extremum-Normalized Stochastic Process Theory (ENSPT), a general framework for representing a continuous stochastic trajectory by its realized extremum geometry and a normalized stochastic shape. Let $S = (S_t)_{0 \leq t \leq 1}$ be a continuous process, let H and L denote its realized high and low, and let τ_H and τ_L denote corresponding extremum times. The principal processes are

$$h_t = \frac{S_t - t}{H - \tau_H}, \quad \ell_t = \frac{S_t - t}{L - \tau_L}. \quad (1)$$

They satisfy the exact reconstruction identities

$$(H - \tau_H)h_t = (L - \tau_L)\ell_t = S_t - t \quad (2)$$

and

$$S_t = t + \frac{(H - \tau_H)h_t + (L - \tau_L)\ell_t}{2}. \quad (3)$$

The theory is developed as a path-space construction, with particular attention to Brownian motion, geometric Brownian motion, barrier-conditioned diffusions, extremum geometry, functional statistics, financial econometrics, simulation, machine learning, and interdisciplinary interpretation. The treatment distinguishes exact mathematical consequences from empirical hypotheses and from metaphorical extensions.

The paper ends with “The End”

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1 Foundations

Let

$$S = (S_t)_{0 \leq t \leq 1} \quad (4)$$

be a continuous stochastic process defined on

$$(\Omega, \mathcal{F}, \mathbb{P}). \quad (5)$$

The interval $[0, 1]$ represents one normalized observation horizon, such as a trading day. Define

$$H = \max_{0 \leq u \leq 1} S_u, \quad (6)$$

$$L = \min_{0 \leq u \leq 1} S_u. \quad (7)$$

Let

$$\tau_H \in \arg \max_{0 \leq u \leq 1} S_u, \quad (8)$$

$$\tau_L \in \arg \min_{0 \leq u \leq 1} S_u. \quad (9)$$

The central random object is

$$\mathcal{G} = (H, L, \tau_H, \tau_L), \quad (10)$$

which will be called the *extremum geometry*.

Define the deterministic-reference-transformed process

$$X_t = S_t - t. \quad (11)$$

Whenever

$$(H - \tau_H)(L - \tau_L) \neq 0, \quad (12)$$

define

$$h_t = \frac{X_t}{H - \tau_H}, \quad (13)$$

$$\ell_t = \frac{X_t}{L - \tau_L}. \quad (14)$$

2 The Fundamental Algebra

Theorem 2.1 (Exact reconstruction). *For every realized path for which the normalizations are defined,*

$$(H - \tau_H)h_t = (L - \tau_L)\ell_t = S_t - t. \quad (15)$$

Consequently,

$$S_t = t + (H - \tau_H)h_t, \quad (16)$$

$$S_t = t + (L - \tau_L)\ell_t. \quad (17)$$

Hence

$$\boxed{S_t = t + \frac{(H - \tau_H)h_t + (L - \tau_L)\ell_t}{2}}. \quad (18)$$

Proof. Direct substitution into the definitions gives

$$(H - \tau_H)h_t = (H - \tau_H) \frac{S_t - t}{H - \tau_H} = S_t - t. \quad (19)$$

The low-normalized equality is identical. Averaging the two exact representations proves the final identity. $\square$

Define

$$A_H = H - \tau_H, \quad A_L = L - \tau_L. \quad (20)$$

Then

$$S_t = t + A_H h_t = t + A_L \ell_t. \quad (21)$$

Corollary 2.2 (Extremum-ratio representation). *Define*

$$R = \frac{A_L}{A_H} = \frac{L - \tau_L}{H - \tau_H}. \quad (22)$$

Then

$$\boxed{h_t = R \ell_t.} \quad (23)$$

Thus h and ℓ are two normalizations of one underlying stochastic trajectory rather than two independent stochastic factors.

3 Extremum-Normalized Shape

The canonical normalized process is

$$\boxed{Z_t = \frac{S_t - t}{H - \tau_H}.} \quad (24)$$

Then

$$Z_t = h_t \quad (25)$$

and

$$S_t = t + (H - \tau_H) Z_t. \quad (26)$$

At the high time,

$$Z_{\tau_H} = 1. \quad (27)$$

If

$$A_H > 0, \quad (28)$$

the extremal inequalities

$$L \leq S_t \leq H \quad (29)$$

become

$$\frac{L - t}{A_H} \leq Z_t \leq \frac{H - t}{A_H}. \quad (30)$$

The corridor width is

$$\frac{H - L}{A_H}, \quad (31)$$

which is constant in t .

Define

$$Y_t = Z_t + \frac{t}{A_H}. \quad (32)$$

Then

$$Y_t = \frac{S_t}{A_H} \quad (33)$$

and

$$\frac{L}{A_H} \leq Y_t \leq \frac{H}{A_H}. \quad (34)$$

Thus the apparently moving normalized corridor becomes a fixed random interval after an affine transformation.

4 Path Space and Information

The process Z is naturally a random element of

$$C[0, 1]. \quad (35)$$

The extremum variables are generally future-dependent. In the original filtration,

$$H \notin \mathcal{F}_t \quad (36)$$

for $t < 1$ in general.

Define

$$\mathcal{E} = \sigma(H, L, \tau_H, \tau_L) \quad (37)$$

and

$$\mathcal{G}_t = \mathcal{F}_t \vee \mathcal{E}. \quad (38)$$

Then the normalization constants are $\mathcal{G}_0$ -measurable and Z is $\mathcal{G}_t$ -adapted.

This establishes the distinction

$$\boxed{\text{ex-post path normalization} \neq \text{ex-ante observable signal.}} \quad (39)$$

This distinction is essential in financial econometrics because use of eventual extrema in a forecasting model would generate look-ahead bias.

5 General Diffusion Theory

Suppose

$$dS_t = \mu_t dt + \sigma_t dW_t. \quad (40)$$

Then

$$dX_t = (\mu_t - 1)dt + \sigma_t dW_t. \quad (41)$$

Conditional on the realized extremum geometry,

$$dZ_t = \frac{\mu_t - 1}{A_H} dt + \frac{\sigma_t}{A_H} dW_t. \quad (42)$$

Consequently,

$$[Z]_t = \frac{[S]_t}{A_H^2}. \quad (43)$$

For the low-normalized process,

$$[\ell]_t = \frac{[S]_t}{A_L^2}. \quad (44)$$

Their cross-variation is

$$[Z, \ell]_t = \frac{[S]_t}{A_H A_L}. \quad (45)$$

The conditional covariance matrix is therefore

$$\Sigma_t = \sigma_t^2 \begin{pmatrix} A_H^{-2} & (A_H A_L)^{-1} \\ (A_H A_L)^{-1} & A_L^{-2} \end{pmatrix}, \quad (46)$$

with

$$\det(\Sigma_t) = 0. \quad (47)$$

Thus the normalized pair has stochastic rank one.

6 Brownian Motion

Consider

$$dS_t = \mu dt + \sigma dW_t. \quad (48)$$

Then

$$S_t = s_0 + \mu t + \sigma W_t \quad (49)$$

and

$$X_t = s_0 + (\mu - 1)t + \sigma W_t. \quad (50)$$

The distinction between the process defining the extrema and the process being normalized is important:

$$\tau_H = \arg \max_t S_t \quad (51)$$

need not equal

$$\arg \max_t (S_t - t). \quad (52)$$

6.1 Killed Kernel

For $\ell < h$, let

$$p_{\ell,h}(t; x, y) \quad (53)$$

be the transition density of the diffusion killed at ℓ and h .

Its spectral representation is

$$p_{\ell,h}(t; x, y) = \frac{2}{h - \ell} e^{\frac{\mu(y-x)}{\sigma^2} - \frac{\mu^2 t}{2\sigma^2}} \sum_{n=1}^{\infty} e^{-\frac{n^2 \pi^2 \sigma^2 t}{2(h-\ell)^2}} \sin\left(\frac{n\pi(x-\ell)}{h-\ell}\right) \sin\left(\frac{n\pi(y-\ell)}{h-\ell}\right). \quad (54)$$

The zero-drift version is

$$p_{\ell,h}(t; x, y) = \frac{2}{h - \ell} \sum_{n=1}^{\infty} e^{-\frac{n^2 \pi^2 \sigma^2 t}{2(h-\ell)^2}} \sin\left(\frac{n\pi(x-\ell)}{h-\ell}\right) \sin\left(\frac{n\pi(y-\ell)}{h-\ell}\right). \quad (55)$$

These kernels are standard in first-passage and diffusion theory [6, 4, 8, 11].

6.2 Conditioned Bridges

For $a < t < b$, conditional on

$$S_a = x, \quad S_b = y, \quad (56)$$

and survival inside (ℓ, h) ,

$$\mathbb{P}(S_t \in dz \mid S_a = x, S_b = y, \ell < S_u < h) = \frac{p_{\ell,h}(t-a; x, z) p_{\ell,h}(b-t; z, y)}{p_{\ell,h}(b-a; x, y)} dz. \quad (57)$$

The extremum-conditioned trajectory is consequently a boundary-constrained bridge rather than an unconstrained Brownian bridge.

7 Extremum Geometry

Define

$$\mathcal{G} = (H, L, \tau_H, \tau_L). \quad (58)$$

If

$$\tau_L < \tau_H, \quad (59)$$

the path decomposes into

$$[0, \tau_L], \quad [\tau_L, \tau_H], \quad [\tau_H, 1]. \quad (60)$$

The corresponding probability structure is represented by

$$\boxed{\text{initial-to-low flux} \times \text{low-to-high excursion} \times \text{high-to-terminal flux.}} \quad (61)$$

If

$$\tau_H < \tau_L, \quad (62)$$

the analogous structure is

$$\boxed{\text{initial-to-high flux} \times \text{high-to-low excursion} \times \text{low-to-terminal flux.}} \quad (63)$$

The probability current for a diffusion density p is

$$J = \mu p - \frac{\sigma^2}{2} \frac{\partial p}{\partial x}. \quad (64)$$

At absorbing boundaries, first-passage probabilities are determined by the appropriate normal derivative of the killed kernel [4, 11].

A convenient regularized construction is to replace (L, H) by

$$(L + \varepsilon, H - \varepsilon) \quad (65)$$

and take the limit

$$\varepsilon \downarrow 0. \quad (66)$$

This avoids treating a boundary-to-boundary transition density as an ordinary interior transition density.

8 Extremum-Ratio Distribution

Let

$$A_H = H - \tau_H, \quad A_L = L - \tau_L, \quad (67)$$

and

$$R = \frac{A_L}{A_H}. \quad (68)$$

The transformation

$$(A_H, A_L) \longrightarrow (A_H, R) \quad (69)$$

has Jacobian

$$\left| \frac{\partial(A_H, A_L)}{\partial(A_H, R)} \right| = |A_H|. \quad (70)$$

Therefore, whenever a joint density exists,

$$f_R(r) = \int \int \int |a| f_{A_H, A_L, \tau_H, \tau_L}(a, ra, \tau, \lambda) da d\tau d\lambda. \quad (71)$$

The variable R completely determines the proportionality between the two normalized processes:

$$h_t = R\ell_t. \quad (72)$$

9 Geometric Brownian Motion

Consider

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (73)$$

Set

$$Y_t = \log S_t. \quad (74)$$

Itô's lemma gives

$$dY_t = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t. \quad (75)$$

Define

$$M = \max_t Y_t, \quad (76)$$

$$m = \min_t Y_t. \quad (77)$$

Then

$$H = e^M, \quad (78)$$

$$L = e^m. \quad (79)$$

Because the logarithm is monotone,

$$\tau_H = \arg \max_t Y_t, \quad (80)$$

$$\tau_L = \arg \min_t Y_t. \quad (81)$$

Thus the GBM extremum problem reduces to drifted Brownian extremum geometry in logarithmic coordinates.

The density transformation is

$$f_{L,H,\tau_L,\tau_H}(\ell, h, \lambda, \tau) = \frac{1}{\ell h} f_{m,M,\tau_L,\tau_H}(\log \ell, \log h, \lambda, \tau). \quad (82)$$

The original ENSPT process remains

$$Z_t = \frac{e^{Y_t} - t}{e^M - \tau_H}. \quad (83)$$

Applying Itô's lemma directly to Z yields

$$dZ_t = \frac{\mu S_t - 1}{H - \tau_H} dt + \frac{\sigma S_t}{H - \tau_H} dW_t. \quad (84)$$

Its quadratic variation is

$$[Z]_t = \frac{\sigma^2}{(H - \tau_H)^2} \int_0^t S_u^2 du. \quad (85)$$

10 The Conditional and Unconditional Laws

The conditional normalized path law is

$$\mathbb{P}_Z(dz \mid \mathcal{G} = g). \quad (86)$$

The unconditional law is

$$\boxed{\mathbb{P}_Z(dz) = \int \mathbb{P}_Z(dz \mid \mathcal{G} = g) \mathbb{P}_{\mathcal{G}}(dg).} \quad (87)$$

For every measurable functional

$$F : C[0, 1] \rightarrow \mathbb{R}, \quad (88)$$

$$\mathbb{E}[F(Z)] = \int \mathbb{E}[F(Z) \mid \mathcal{G} = g] \mathbb{P}_{\mathcal{G}}(dg). \quad (89)$$

Hence

$$\text{Var}(F(Z)) = \mathbb{E}[\text{Var}(F(Z) \mid \mathcal{G})] + \text{Var}[\mathbb{E}(F(Z) \mid \mathcal{G})]. \quad (90)$$

This is the central hierarchical decomposition:

total shape uncertainty = conditional path uncertainty + extremum-geometry uncertainty.

(91)

11 Finance and Economics

For financial applications, S_t may represent a stock price, index level, exchange rate, commodity price, or another tradable state variable.

For trading day d ,

$$Z_{d,t} = \frac{S_{d,t} - t}{H_d - \tau_{H,d}}. \quad (92)$$

The daily geometry is

$$\mathcal{G}_d = (H_d, L_d, \tau_{H,d}, \tau_{L,d}). \quad (93)$$

This separates:

$$\text{price scale,} \quad \text{extremum geometry,} \quad \text{normalized path shape.} \quad (94)$$

Potential economic applications include comparative analysis of intraday market structure, volatility normalization, regime classification, and cross-asset functional representations.

The construction does not by itself imply arbitrage opportunities or forecastability. Standard no-arbitrage theory, equivalent martingale measures, and market completeness remain separate issues [3, 9, 12].

12 Econometrics

For N observed trading days,

$$\{Z_{d,t} : d = 1, \dots, N, t \in [0, 1]\} \quad (95)$$

forms functional panel data.

The empirical mean is

$$\hat{m}_Z(t) = \frac{1}{N} \sum_{d=1}^N Z_{d,t}. \quad (96)$$

The empirical covariance function is

$$\hat{K}_Z(s, t) = \frac{1}{N-1} \sum_{d=1}^N (Z_{d,s} - \hat{m}_Z(s)) (Z_{d,t} - \hat{m}_Z(t)). \quad (97)$$

A functional regression may be specified as

$$Y_d = \alpha + \int_0^1 \beta(t) Z_{d,t} dt + \gamma^\top X_d + \varepsilon_d. \quad (98)$$

The extremum geometry can enter separately through

$$X_d^{\text{ext}} = (H_d, L_d, \tau_{H,d}, \tau_{L,d}, R_d). \quad (99)$$

This allows shape effects and extremum effects to be statistically separated.

However, ex-post extrema must not be used as predictors for information sets that precede those extrema.

13 Functional Statistics

Define

$$m_Z(t) = \mathbb{E}[Z_t] \quad (100)$$

and

$$K_Z(s, t) = \mathbb{E}[(Z_s - m_Z(s))(Z_t - m_Z(t))]. \quad (101)$$

The covariance operator is

$$(\mathcal{K}f)(t) = \int_0^1 K_Z(s, t)f(s)ds. \quad (102)$$

The eigenvalue problem is

$$\mathcal{K}\phi_k = \lambda_k\phi_k. \quad (103)$$

The Karhunen–Loève representation is

$$Z_t = m_Z(t) + \sum_{k=1}^{\infty} \xi_k \phi_k(t), \quad (104)$$

with

$$\mathbb{E}[\xi_k] = 0, \quad \mathbb{E}[\xi_j \xi_k] = \lambda_k \mathbf{1}_{\{j=k\}}. \quad (105)$$

This provides a principled dimensionality reduction for normalized intraday trajectories [10, 7].

14 Statistics of Realized Volatility

For an Itô diffusion,

$$[Z]_1 = \frac{1}{(H - \tau_H)^2} [S]_1. \quad (106)$$

For GBM,

$$[Z]_1 = \frac{\sigma^2}{(H - \tau_H)^2} \int_0^1 S_t^2 dt. \quad (107)$$

Consequently, ENSPT creates a natural decomposition between realized volatility and realized extremum scale.

High-frequency econometric implementation must account for market microstructure noise, irregular sampling, jumps, and discretization [1, 2, 5].

15 Physics and Scaling

The representation

$$S_t = t + A_H Z_t \quad (108)$$

has the mathematical structure of a scale–shape decomposition.

The dimensionless range statistic

$$\frac{H - L}{\sigma} \quad (109)$$

and extremum ratio

$$R = \frac{L - \tau_L}{H - \tau_H} \quad (110)$$

are natural similarity variables.

The analogy to scaling theory and nondimensionalization in physics is structural rather than ontological; ENSPT does not assert that financial markets obey physical laws.

16 Biology and Dynamical Systems

For a biological state variable X_t , choose a deterministic reference trajectory $c(t)$ and define

$$M = \max_t X_t, \quad \tau_M = \arg \max_t X_t. \quad (111)$$

Then

$$Z_t^X = \frac{X_t - c(t)}{M - c(\tau_M)}. \quad (112)$$

The same mathematical representation may be applied to growth curves, physiological measurements, ecological trajectories, and other continuous dynamical systems.

Such applications require domain-specific measurement models and should not be inferred merely from the mathematical analogy.

17 Psychology and Psychiatry

Longitudinal psychological observations can be represented as stochastic trajectories. An extremum-normalized representation could compare trajectory shape after removing an endogenous scale.

For example,

$$Z_t^P = \frac{P_t - c(t)}{P_{\max} - c(\tau_{\max})}. \quad (113)$$

The framework is descriptive and mathematical. It does not provide a diagnostic criterion, clinical intervention, or psychiatric causal theory.

18 Philosophy

The framework distinguishes:

$$\text{path realization} \quad (114)$$

from

$$\text{information-conditioned path description.} \quad (115)$$

The conditional law

$$\mathbb{P}(Z. \mid \mathcal{G}) \quad (116)$$

formalizes the idea that the description of a trajectory depends on what information is treated as known.

The distinction between ex ante and ex post information is therefore not merely a philosophical metaphor; it is represented mathematically through filtrations and conditional probability.

19 Religion and Interpretive Thought

ENSPT admits metaphorical interpretations involving trajectory, boundary, transformation, and perspective. Such interpretations may be useful in philosophy of religion or comparative intellectual history.

They do not constitute mathematical consequences of the stochastic theory.

The mathematical content remains:

$$S_t = t + (H - \tau_H)Z_t. \quad (117)$$

20 Political Science and International Relations

For an aggregate institutional, economic, demographic, or capability index C_t , define

$$Z_t^C = \frac{C_t - c(t)}{C_{\max} - c(\tau_{\max})}. \quad (118)$$

This can facilitate descriptive comparison of trajectory shapes across countries or institutions with different absolute scales.

The framework does not determine political preferences, strategic intentions, causal mechanisms, or diplomatic outcomes. Those require institutional and historical evidence.

21 Welfare Economics

For a welfare index W_t ,

$$Z_t^W = \frac{W_t - c(t)}{W_{\max} - c(\tau_{\max})}. \quad (119)$$

This separates a scale component from a normalized welfare trajectory.

The construction does not solve interpersonal utility comparison, aggregation, or normative welfare evaluation. Those remain substantive questions in welfare economics.

22 Warfare Economics and Strategic Resources

For an abstract resource-capacity trajectory Q_t ,

$$Z_t^Q = \frac{Q_t - c(t)}{Q_{\max} - c(\tau_{\max})}. \quad (120)$$

The construction can describe temporal resource shape independently of one realized scale.

It is a descriptive mathematical framework and does not provide operational military targeting or tactical instructions.

23 Computer Science and Algorithms

A deterministic implementation for an observed path is:

INPUT:

```
time grid t[0], ..., t[n]
trajectory S[0], ..., S[n]
```

```
H      = maximum(S)
L      = minimum(S)
```

```

tau_H    = time(H)
tau_L    = time(L)

A_H      = H - tau_H
A_L      = L - tau_L

FOR each observation i:
    h[i] = (S[i] - t[i]) / A_H
    l[i] = (S[i] - t[i]) / A_L
    Z[i] = h[i]
    reconstruction[i] =
        t[i] + (A_H*h[i] + A_L*l[i])/2
    residual[i] = reconstruction[i] - S[i]

R = A_L / A_H

RETURN:
    H, L, tau_H, tau_L,
    A_H, A_L, R,
    Z, l, residual

```

In exact arithmetic,

$$\text{residual}[i] = 0. \quad (121)$$

Nonzero residuals diagnose implementation, sampling, or measurement issues.

24 Simulation Algorithm

A model-based simulation is:

```

INPUT:
    stochastic model for S
    number of paths N
    time grid

FOR d = 1,...,N:

    simulate S_d

    H_d      = max(S_d)
    L_d      = min(S_d)
    tau_H_d  = argmax(S_d)
    tau_L_d  = argmin(S_d)

    A_H_d = H_d - tau_H_d
    A_L_d = L_d - tau_L_d

    IF A_H_d != 0 AND A_L_d != 0:

        FOR each t:
            Z_d[t] = (S_d[t] - t) / A_H_d
            l_d[t] = (S_d[t] - t) / A_L_d

```

```

R_d = A_L_d / A_H_d

END IF

END FOR

Estimate:
  extremum geometry distribution
  distribution of R
  mean normalized path
  covariance function
  functional principal components
  model-data discrepancies

```

25 Data Science and AI/ML

The functional trajectory can be represented through principal-component scores:

$$Z_{d,t} \approx m_Z(t) + \sum_{k=1}^K \xi_{dk} \phi_k(t). \quad (122)$$

A compact feature vector is therefore

$$\mathbf{x}_d = (H_d, L_d, \tau_{H,d}, \tau_{L,d}, R_d, \xi_{d1}, \dots, \xi_{dK}). \quad (123)$$

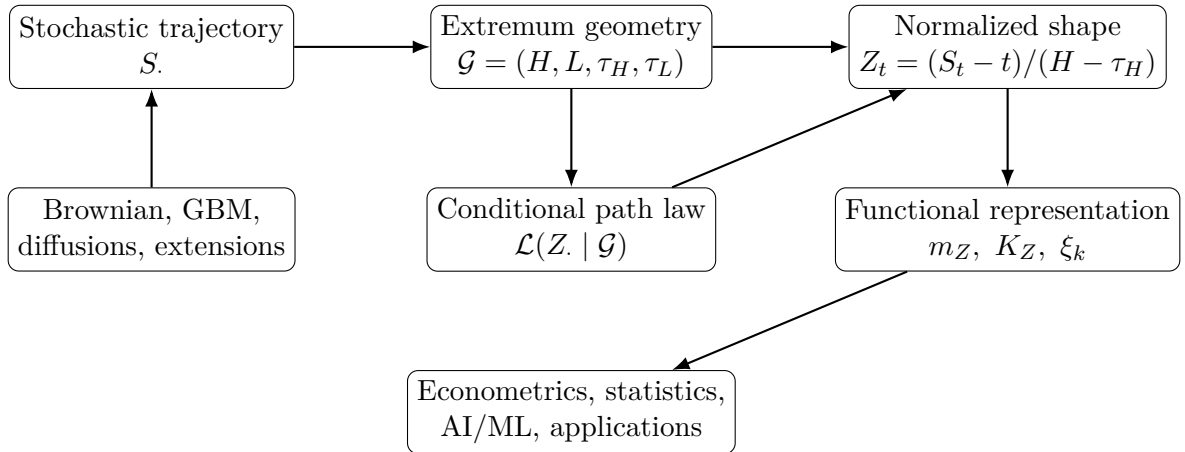
A generic predictive or classification architecture may be written

$$\hat{Y}_d = f_\theta(\mathbf{x}_d). \quad (124)$$

Possible models include linear functional regression, Gaussian processes, kernel methods, tree ensembles, recurrent networks, state-space models, transformers, and other sequence architectures.

The normalization stage must remain separated from the prediction stage when future information would otherwise leak into the features.

26 Unified Architecture



The theory therefore has the hierarchy

$$\boxed{\begin{matrix} \textit{stochastic} & \rightarrow & \textit{extremum} & \rightarrow & \textit{conditioned} & \rightarrow & \textit{normalized} & \rightarrow & \textit{statistical} \\ \textit{dynamics} & & \textit{geometry} & & \textit{path} & & \textit{shape} & & \textit{representation} \end{matrix}} \quad (125)$$

27 Interdisciplinary Map

Field	Principal ENSPT object	Role
Stochastic calculus	$S_t, [S]_t, \mathcal{G}$	Diffusion, filtration, quadratic variation, conditioning.
Probability	$\mathbb{P}(Z. \mathcal{G})$	Conditional and unconditional path laws.
Mathematical finance	Z_t, H, L	Scale-normalized asset trajectories.
Econometrics	$Z_{d,t}$	Functional panel representation and model comparison.
Statistics	m_Z, K_Z, ϕ_k, ξ_k	Functional inference and dimensional reduction.
Physics	$S_t = t + A_H Z_t$	Scale–shape and dimensionless representation.
Biology	Z_t^X	Normalized dynamical trajectories.
Psychology	Z_t^P	Descriptive longitudinal trajectory comparison.
Psychiatry	Z_t^P	Potential trajectory representation, not diagnosis.
Philosophy	$\mathbb{P}(Z \mathcal{G})$	Ex ante/ex post information distinction.
Religion	Shape–scale interpretation	Metaphorical rather than mathematical.
Political science	Z_t^C	Descriptive institutional trajectory normalization.
Geopolitics	$\mathcal{G}$	Descriptive aggregate trajectory geometry.
Diplomacy	Path-conditioned information	Formal analogy concerning information and perspective.
International relations	Z_t^C	Cross-unit trajectory comparison.
Welfare economics	Z_t^W	Normalized welfare trajectories.
Warfare economics	Z_t^Q	Abstract resource-capacity trajectories.
Computer science	Algorithmic construction	Deterministic extraction and validation.
Data science	Functional observations	Representation, clustering, anomaly detection.
AI/ML	$\mathbf{x}_d$	Low-dimensional trajectory features.
Control theory	Conditioned diffusion	Potential state normalization and trajectory analysis.
Information theory	$\mathcal{G}$ and $Z.$	Decomposition of path information into geometry and shape.

28 Statistical Testing Program

Suppose empirical normalized paths are

$$Z_1, \dots, Z_N \quad (126)$$

and a theoretical model generates

$$Z_1^{(0)}, \dots, Z_M^{(0)}. \quad (127)$$

One may compare the empirical and theoretical distributions using an appropriate functional distance:

$$D = d\left(\widehat{\mathbb{P}}_Z, \mathbb{P}_Z^{(0)}\right). \quad (128)$$

Possible choices include integrated squared distance, energy distance, kernel discrepancies, Wasserstein distance on discretized paths, or likelihood-based criteria when a tractable likelihood exists.

The null hypothesis may be expressed as

$$H_0 : \mathbb{P}_Z^{\text{emp}} = \mathbb{P}_Z^{\text{model}}. \quad (129)$$

The analysis should separately test:

1. the extremum-geometry distribution;
2. the conditional path-shape distribution;
3. the unconditional normalized-path distribution.

This prevents a model from appearing successful merely because it reproduces one component of the distribution.

29 Extensions

The framework naturally extends beyond Brownian and GBM models.

A stochastic-volatility model may be written

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \quad (130)$$

$$d\sigma_t = a(\sigma_t, t)dt + b(\sigma_t, t)dB_t. \quad (131)$$

A jump-diffusion extension is

$$dS_t = \mu_t dt + \sigma_t dW_t + \int_{\mathbb{R}} x \tilde{N}(dt, dx). \quad (132)$$

For jump processes, extrema may be achieved through discontinuous jumps rather than continuous boundary contact. The continuous boundary-flux theory must then be replaced by jump-crossing and overshoot theory.

Multivariate extensions may consider

$$S_t \in \mathbb{R}^d \quad (133)$$

and replace scalar extrema by coordinatewise, radial, support-function, or convex-set extrema.

30 Limitations

The theory has several limitations.

1. The normalizations require nonzero denominators.
2. Discrete data can produce tied or interval-valued extrema.
3. Observed high and low prices need not equal continuous-path extrema.

4. High-frequency observations contain microstructure noise.
5. Ex-post normalization contains future information.
6. Brownian and GBM models are benchmark models, not empirical laws of financial markets.
7. Interdisciplinary applications require independent domain-specific validation.

In particular, ENSPT should not be interpreted as proving that normalized financial paths are universal across assets or markets.

31 Central Theorem

Theorem 31.1 (Extremum-Normalized Stochastic Decomposition). *Let S be a continuous stochastic process on $[0, 1]$ with well-defined extrema H, L and corresponding times τ_H, τ_L . Assume*

$$(H - \tau_H)(L - \tau_L) \neq 0. \quad (134)$$

Define

$$Z_t = \frac{S_t - t}{H - \tau_H} \quad (135)$$

and

$$R = \frac{L - \tau_L}{H - \tau_H}. \quad (136)$$

Then

$$S_t = t + (H - \tau_H)Z_t, \quad (137)$$

$$\ell_t = \frac{Z_t}{R}, \quad (138)$$

$$h_t = Z_t. \quad (139)$$

Furthermore, if

$$\mathcal{G} = (H, L, \tau_H, \tau_L), \quad (140)$$

then the law of Z admits the mixture representation

$$\mathbb{P}_Z(dz) = \int \mathbb{P}_Z(dz \mid \mathcal{G} = g) \mathbb{P}_{\mathcal{G}}(dg). \quad (141)$$

Proof. The first three identities follow directly from the definitions. The mixture representation is the law of total probability applied to the random element Z and the sigma-field generated by $\mathcal{G}$. $\square$

32 Research Agenda

The complete program suggested by ENSPT is:

1. derive and numerically evaluate extremum-geometry laws for selected continuous diffusions;
2. develop accurate simulation of paths conditional on extrema;
3. estimate empirical extremum geometry from high-frequency data;

4. estimate the functional law of normalized paths;
5. compare Brownian, GBM, stochastic-volatility, and jump models;
6. develop leakage-free online approximations;
7. investigate cross-asset universality and non-universality;
8. study regime dependence of normalized path distributions;
9. integrate functional representations with modern machine learning;
10. generalize the construction to multidimensional and discontinuous stochastic processes.

33 Conclusion

Extremum-Normalized Stochastic Process Theory begins with the simple definitions

$$h_t = \frac{S_t - t}{H - \tau_H}, \quad (142)$$

$$\ell_t = \frac{S_t - t}{L - \tau_L}, \quad (143)$$

but yields a complete path-space representation.

The fundamental identity is

$$\boxed{(H - \tau_H)h_t = (L - \tau_L)\ell_t = S_t - t.} \quad (144)$$

Consequently,

$$\boxed{S_t = t + (H - \tau_H)Z_t, \quad Z_t = h_t.} \quad (145)$$

The theory separates the stochastic trajectory into

$$\boxed{\text{extremum geometry} + \text{normalized stochastic shape}.} \quad (146)$$

For Brownian motion, the relevant conditional laws are generated by killed diffusions, boundary fluxes, excursion limits, and conditioned bridges. For geometric Brownian motion, logarithmic transformation reduces the extremum geometry to drifted Brownian motion, followed by a nonlinear transformation back into price space.

The complete probabilistic object is therefore

$$\boxed{\mathfrak{Z} = (H, L, \tau_H, \tau_L, Z.),} \quad (147)$$

with law

$$\boxed{\mathbb{P}_{\mathfrak{Z}} = \mathbb{P}_{\mathcal{G}}\mathbb{P}_{Z|\mathcal{G}}.} \quad (148)$$

The resulting framework is simultaneously a stochastic-process construction, a path-space normalization, a functional-statistical representation, and a potential methodology for empirical comparison of stochastic dynamical models. Its strongest mathematical statement is not a forecasting claim but an exact representation of a completed trajectory relative to its own extremal geometry.

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Glossary

Extremum geometry

The random vector

$$\mathcal{G} = (H, L, \tau_H, \tau_L). \quad (149)$$

Extremum-normalized stochastic process

A stochastic trajectory divided by a scale determined by a realized extremum and its occurrence time.

Normalized shape

The path

$$Z_t = \frac{S_t - t}{H - \tau_H}. \quad (150)$$

Extremum ratio

The random variable

$$R = \frac{L - \tau_L}{H - \tau_H}. \quad (151)$$

Killed diffusion

A diffusion terminated upon reaching a specified boundary.

Boundary flux

Probability flow through an absorbing boundary, used in first-passage analysis.

Bridge

A stochastic process conditioned on endpoint information.

Doob transform

A transformation producing a conditioned Markov process from a suitable positive harmonic or excessive function.

Quadratic variation

The accumulated limiting squared increments of a stochastic process.

Functional data

Data whose observations are curves or functions.

Karhunen–Loève expansion

An eigenfunction representation of a second-order stochastic process based on its covariance operator.

Look-ahead bias

Statistical bias created by using information unavailable at the decision time.

Geometric Brownian motion

The diffusion

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (152)$$

Path space

The set of possible trajectories, here typically $C[0, 1]$.

Shape–scale decomposition

The representation

$$S_t = t + A_H Z_t. \quad (153)$$

Extremum-conditioned law

The conditional probability law

$$\mathbb{P}(Z. \mid H, L, \tau_H, \tau_L). \quad (154)$$

The End